1 Proposal

Title presentation: the problem, the solution and the project name.

PROJECT

Title	USWAG: Unified System for Work and Growth
Proponents	Bryan Jay Lumabas and Claire Pillo
Client	Samahan ng Masisinop, Sustainable Livelihood Program (SLP), Barangay Aniog

- USWAG is a relational database that tracks the stock, sales, receipts and expenses of SLP microenterprises.
- Triggers keep stock and seller capital correct without help from any application.

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THE PROBLEM

- Producers keep stock, sales and costs in notebooks, so mistakes stay hidden until money runs short.
- Stock and capital drift apart because nothing checks them.
- There is no proof of payment, audit trail or cash-flow view for SLP admins.
- Rural signal is weak, so sales must be recordable offline.

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THE SOLUTION

- One normalized database (12 tables, BCNF) is the single source of truth.
- Triggers and procedures enforce the rules, so the data protects itself.
- Offline sales sync without duplicates, and every paid sale gets a JSON receipt.
- Verified inventory and an append-only capital ledger give a full audit trail.

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OBJECTIVE AND WHY IT MATTERS

- Objective: design and prove a rule-enforcing database through modeling, normalization, DDL, triggers, queries and indexing.
- A spreadsheet cannot refuse a bad entry, but the database rejects negative stock, edited balances and empty paid sales.
- Every peso must be traceable, and any interface can change while the database stays reliable.

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2 Modeling

Conceptual model: ERD and business rules in Crow’s Foot notation.

ENTITIES AND CARDINALITY

- Task: model the domain in Crow’s Foot notation with exact cardinalities.
- Result: 12 entities and 16 relationships, generated from the live database.
- Strong entities: Accounts, Sellers, Categories, Products and Sales.
- Weak entities: Sale_Details (identified by Sales) and Capital_Ledger (identified by Sellers).
- Sale_Details is the associative entity that resolves Sales and Products and carries quantity_sold and unit_price_at_sale.
- Optional links (0..1): the buyer on a sale and the verifier on an inventory entry.

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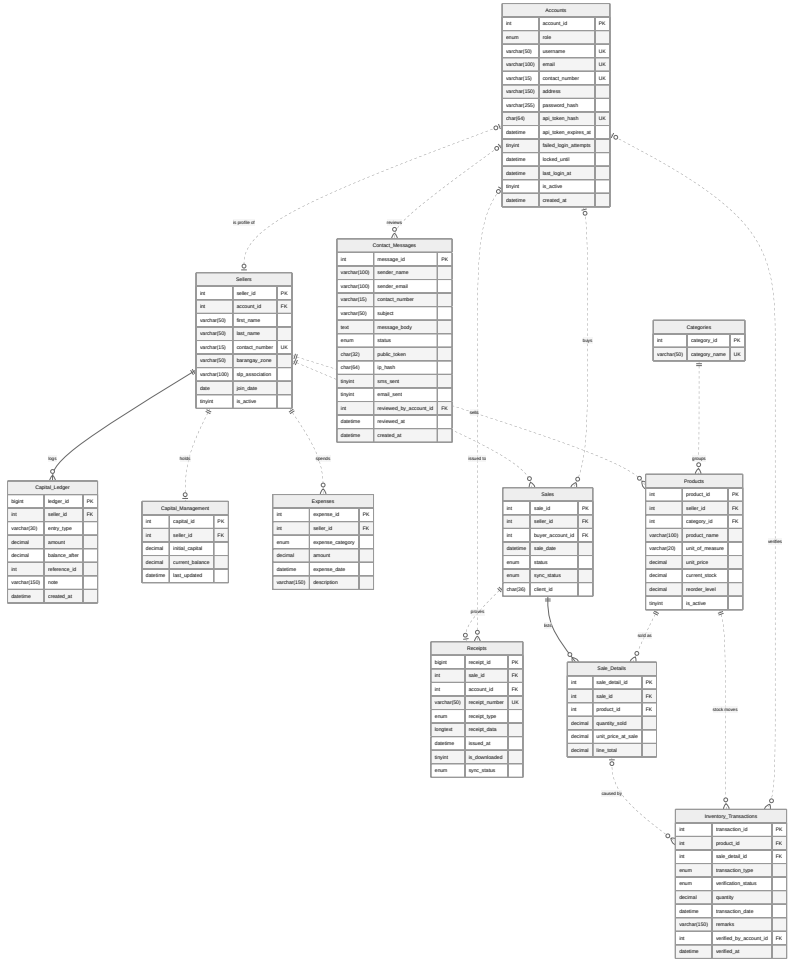
BUSINESS RULES ENFORCED BY THE DATABASE

- Stock changes only through admin-verified inventory entries.
- A paid sale credits capital once and issues one receipt, and an empty sale cannot be paid.
- Expenses debit capital, and every money movement is written to the ledger.
- Offline sales carry a client UUID, so syncing never creates duplicates.

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2 · Modeling

ENTITY-RELATIONSHIP DIAGRAM



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3 Normalization

Formal normalization: from one unnormalized form to BCNF.

DEPENDENCIES AND KEY

- sale_id → seller, buyer, date, status.
- seller_id → name, zone; product_id → name, category, price; category_id → category_name.
- (sale_id, product_id) → quantity_sold, unit_price_at_sale.
- The candidate key of the unnormalized form is {sale_id, product_id}.

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3 · Normalization

DECOMPOSITION

UNF
A sale form with a repeating group of products.

1NF
One row per sale and product, with atomic values.

2NF
Partial dependencies split into Sales, Products and Sale_Details.

3NF
Transitive dependencies split into Sellers, Accounts and Categories.

BCNF
Every determinant is a superkey, checked on all 12 tables.

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3 · Normalization

PROOFS

- Lossless: each split joins on the key of one side (Heath’s theorem).
- Dependency-preserving: every dependency sits inside one final table.
- unit_price_at_sale is an intended price snapshot that depends on the sale line, so it is not a violation.

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4 Implementation

Physical schema: executable SQL that enforces integrity.

DATABASE STRUCTURE GALLERY

Accounts

```
CREATE TABLE IF NOT EXISTS Accounts (
    account_id INT PRIMARY KEY AUTO_INCREMENT,
    role ENUM('Buyer', 'Seller', 'SLP Admin') NOT NULL DEFAULT 'Buyer',
    username VARCHAR(50) NOT NULL,
    email VARCHAR(100) NULL,   -- optional; unique when set
    contact_number VARCHAR(15) NOT NULL,
    address VARCHAR(150) NULL,
    password_hash VARCHAR(255) NOT NULL,
    api_token_hash CHAR(64) NULL,
    api_token_expires_at DATETIME NULL,
    failed_login_attempts TINYINT UNSIGNED NOT NULL DEFAULT 0,
```

Tables 1 / 12

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STRENGTHS

- The script imports with 0 errors and rebuilds safely: 12 tables, 9 views, 18 procedures and 26 triggers.
- Data protects itself: negative stock is rejected, verified inventory cannot be deleted and line totals are generated.
- Stock and capital update automatically, and the ledger balances: 5,000 + 2,600 − 1,500 = 6,100.
- Offline sync is idempotent, and each paid sale builds an 80 mm JSON receipt.
- It runs on MySQL 8 and MariaDB 10.11, whatever frontend sits on top.

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WEAKNESSES FOUND AND HANDLED

- Fixed: a direct UPDATE could set a balance to 999,999 without a ledger entry, so balances now change only through sp_capital_move.
- Open: registration leaves Sellers.first_name and last_name unfilled, so reports show the username.
- Fixed in the final schema: the lab version created receipts too early, had no buyer foreign key and let CASCADE erase history.
- Trade-off: 26 triggers add write cost and hide logic inside the database.

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4 · Implementation

PERFORMANCE

- Query: cash flow per seller, joined to sale lines.
- Before: a full scan of 100,000 rows took 146 ms.
- After idx_sales_seller_status and idx_saledetails_sale: 250 index probes took 2.5 ms, about 58 times faster.

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5 Deliverables

Proof of completion: view, download or print each file.

SOFTCOPY DOCUMENTS

.SVG **ERD (Crow's Foot)** Live Mermaid diagram of all 12 entities.	.MMD **ERD Mermaid Code** The source that draws the ERD.	.PDF **Business Rules** The 10 rules the schema enforces.
.PDF **Normalization Proofs** Dependencies from UNF to BCNF.	.PDF **Performance Report** Index test: 146 ms to 2.5 ms.	.SQL **uswag_db.sql** Tables, triggers, procedures, views and queries.
.DOCX **Proposal (Word)** The presentation as a Word file.	.PDF **Proposal (PDF)** The presentation as 16:9 PDF slides.	.PPTX **Slides (Plain)** Editable PowerPoint slides.
.PPTX **Slides (Interface)** Screenshots of this site as slides.	.MD **READ.md** What is inside and how to run it.	

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